ZIHO KANG

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RESEARCH INTERESTS

My research interests fall at the intersection of human factors, human-systems engineering, and data analytics. Especially, I develop algorithms to map human visual search patterns with human decision making processes that can be used to evaluate human performance, assess situation awareness, model decision making processes, inform the design of interfaces, and develop novel training methods.

Key words: human factors, human systems engineering, data analytics, human-computer interaction, usability research, eye tracking analytics, decision making, training interventions, simulation.
Applications: aerospace/aviation, weather, virtual reality, Deepwater Horizon explorations.

EDUCATION

- Ph.D., Industrial Engineering, Purdue University, West Lafayette, IN. May 2012.
  Dissertation: Dynamic sequence analysis of a decision making task of multielement target tracking and its usage as a learning method. Advisor: Dr. Steven Landry
- M.S., Industrial Engineering, Purdue University, West Lafayette, IN. Dec. 2006.
  Thesis: An improved optimization model for assigning empty freight cars in a railroad network
  Advisor: Dr. Jean-Philippe Richard.
- B.S., Industrial Engineering, Korea University, Seoul, South Korea. Mar. 2001.

PROFESSIONAL APPOINTMENTS

2014-present	Assistant Professor, School of Industrial and Systems Engineering, Affiliate Professor (by courtesy), Department of Psychology, Faculty member of Data Science and Analytics Program, University of Oklahoma, Norman, OK.
2013-2014	Post-doctoral Researcher, College of Computing and Informatics, Drexel University, Philadelphia, PA. Advisor: Dr. Ellen J. Bass.
2012-2013	Post-doctoral Research Associate, Department of Systems and Information Engineering, University of Virginia, Charlottesville, VA. Advisor: Dr. Ellen J. Bass
2004-2011	Research Assistant and Teaching Assistant, School of Industrial Engineering, Purdue University, West Lafayette, IN.
2003-2004	Engineer, Research and Development of Trading Systems, Korea Stocks and Futures Exchange, Pusan, South Korea.
2001-2002	Engineer, Research and Development of Enterprise Resource Planning Systems, Samsung Data Systems, Seoul, South Korea.
1997-1999	Honorable discharge as KATUSA Sergeant, Education Center, Camp Humphreys, USASA, AREA III, South Korea.

AWARDS AND HONORS

Sep 2017 Received **Best Presentation Award** in *Proceedings of the 2nd International Conference on Bio-engineering for Smart Technologies* (2017 BioSmart), Paris, France.

Oct 2016	Received **Andrew P. Sage Best Transaction Award** from the *IEEE Systems, Man, and Cybernetics Society*. Awarded for the best paper published in the *Transactions in Human-Machine Systems*.
Jan 2015	Received **Junior Faculty Fellowship Award** from the Office of the Vice President for Research at the University of Oklahoma.
Dec 2001	Received Best New Recruit Award from Samsung Data Systems (out of approximately 600 recruited).
May 1999	Received U.S. Army Recommendation Medal.
Nov 1998	Received Golden Tiger Decoration upon passing the Primary Leadership Development Program from the U.S. Army.

INVITED PRESENTATIONS, PANEL DISCUSSIONS, AND EXHIBITIONS (Total: 13)

1. "**Human role in complex systems: Trust in automation, information reliance, and information overload**," *Human Systems Integration Summit*, Mike Monroney Aeronautical Center, Federal Aviation Administration, Apr. 11, 2019, OKC, OK. (presentation)
2. "**Universal Design for Learning (UDL) and multimodal training**," *2018 Interservice/Industry Training, Simulation and Education Conference (I/ITSEC)*, Nov. 26-30, 2018. Orlando, FL. (exhibition supported by Federal Aviation Administration Center of Excellence)
3. "**Characterization of air traffic controllers' visual search patterns and control strategies**," *2018 Interservice/Industry Training, Simulation and Education Conference (I/ITSEC)*, Nov. 26-30, 2018. Orlando, FL. (exhibition supported by Federal Aviation Administration Center of Excellence)
4. "**Advancements of eye movement analysis methodologies for a dynamic task**," *Tech Talk.* Air Force Research Laboratory, Department of Defense, Oct. 18, 2018, Austin, TX. (presentation)
5. "**Human factors research roadmap**," *Center of Excellence for Technical Training and Human Performance Workshop*, Federal Aviation Administration Center of Excellence, Jun. 6, 2018, Norman, OK. (panel)
6. "**Technical training roadmap**," *Center of Excellence for Technical Training and Human Performance Workshop*, Federal Aviation Administration Center of Excellence, Jun. 5, 2018, Norman, OK. (panel)
7. "**Human systems integration: Eye tracking for training**," *Human Systems Integration Summit*, Mike Monroney Aeronautical Center, Federal Aviation Administration, May 23, 2018, OKC, OK. (presentation)
8. "**Improving the mission readiness of employees and safety of the National Airspace System (NAS)**," *Solutions for Operational Aviation Research (SOAR) Q2 Meeting*, Federal Aviation Administration Center of Excellence, Apr. 4, 2018, Philadelphia, PA. (panel)
9. "**Data visualizations in eye tracking research**," *Tobii Pro Research Spotlight Seminar Series*, Tobii Pro North America, Jun. 1, 2017. (webinar)
10. "**Eye tracking data analysis algorithms and applications**," Department of Information and Industrial Engineering at Yonsei University, Jul. 21, 2015, Seoul, S. Korea. (presentation)
11. "**Eye tracking data analysis methods for a tracking task of multiple moving targets**," *Federal Aviation Administration Collaborative Research Meeting*, Civil Aerospace Medical Institute, Mike Monroney Aeronautical Center, Federal Aviation Administration, Oct. 10, 2014, OKC, OK. (presentation)
12. "**Analysis of driving behaviors using eye movements**," *1st Hyundai Global Top Talent Forum*, 2011 World Motor Auto Show, Jan. 10, 2012, Detroit. (presentation)
13. "**Capturing and analyzing visual groupings of multiple moving targets in an aircraft conflict detection task using eye movements**," *Air Traffic Control Association (ATCA) 55th Annual*

Conference and Exposition, Oct. 24-27, 2010, Alexandria, V.A. (exhibition supported by the President of ATCA)

PUBLICATIONS (Total: 55)

Underlined names indicate the students who published papers as Kang's advisees or mentees (total of 8 graduate students).

Peer-reviewed Journal Publications (Total: 15)

1. Mandal, S. and **Kang. Z.** (2018). Using eye movement data visualization to enhance training of air traffic controllers: A dynamic network approach. *Journal of Eye Movement Research, 11*(4), 1-20.
2. Salehi, S., Kiran, R., Jeon, J., & **Kang, Z.** (2018). Developing a cross-disciplinary, scenario-based training approach integrated with eye tracking data collection to enhance situational awareness in offshore oil and gas operations. *Journal of Loss Prevention in the Process Industries*, 56, 78-94.
3. Khamaj, A., and **Kang, Z.** (2018). Usability evaluation of mobile weather hazard alert applications. *Industrial and Systems Engineering Review*, *6*(1), 21-40.
4. **Kang, Z.**, Dragoo, M.R., Yeagle, L.N., Shehab, R.L., Yuan, H., Ding, L., and West, S.G. (2018). Adaptive learning pedagogy of Universal Design for Learning (UDL) for multimodal training. *Journal of Aviation/Aerospace Education and Research, 27*(1), 23-48.
5. Wilson, K.A., Heinselman, P.L., and **Kang, Z.** (2018). Comparing Forecaster Eye Movements during the Warning Decision Process. *Weather and Forecasting*, *33*(2), 501-521.
6. Argyle, E.M., Gourley, J.J., Ling, C., Shehab, R.L., and **Kang, Z.** (2017). Effects of display design on signal detection in flash flood forecasting. *International Journal of Human-Computer Studies*, *99*, 48-56.
7. **Kang, Z.** and Kim, K. (2017). Multimodal perception study on virtual 3D curved textures with vision and touch for interactive multimedia systems. *Multimedia Tools and Applications*, *77(2)*, 1-15.
8. **Kang, Z.**, Mandal, S., Crutchfield, J., Millan, A., and McClung, S.N. (2016) Designs and algorithms to map eye tracking data with dynamic multielement moving objects. *Computational Intelligence and Neuroscience*, *2016*, 1-18. (vol. *2016*: Special issue titled Advances in Eye Tracking Technology: Theory, Algorithms, and Applications)
9. **Kang, Z.** and Morin, T. (2016). Multi-attribute decision making under imperfect information and uncertain outcome of an unfair bidding game. *International Journal of Information Technology & Decision Making*, *15(1)*, 63-81.
10. McClung, S.N. and **Kang, Z.** (2016). Characterization of visual scanning patterns in air traffic control. *Computational Intelligence and Neuroscience*, *2016*, 19-35. (vol. *2016*: Special issue titled Advances in Eye Tracking Technology: Theory, Algorithms, and Applications)
11. Wilson, K.A., Heinselman, P.L., Kuster, C.M., Kingfield, D.M., and **Kang, Z.** (2016). Forecaster performance and workload: Does radar update time matter? *Weather and Forecasting*, *32*, 253-274.
12. Wilson, K.A., Heinselman, P.L., and **Kang, Z.** (2016). Exploring applications of eye tracking in operational meteorology research. *Bulletin of the American Meteorological Society*, *97(11)*, 2019-2025.
13. **Kang, Z.** and Landry, S.J. (2015). An eye movement analysis algorithm for a multi-element target tracking task: Maximum transition-based agglomerative hierarchical clustering. *IEEE Transactions on Human-Machine Systems*, *45(1)*, 13-24. **Received Andrew P. Sage Best Transactions Paper Award**.
14. **Kang, Z.** and Landry, S.J. (2014). Using scanpaths as a learning method for a conflict detection task of multiple target tracking. *Human Factors: The Journal of Human Factors and Ergonomics Society*. *56 (1)*, 1150-1162.
15. **Kang, Z.** and Landry, S.J. (2014). Top-down approach for a linguistic fuzzy logic model. *Cybernetics and Systems: An International Journal*, *45(1)*, 39-55.

Peer-reviewed Journal Publications Under Review

1. Salehi, S., Naqvi S. A. M., Raza, M., and Kang, Z. (Submitted: Jan, 2019). Process Tracing Through Eye Tracking in Simulation-Based Training to Enhance Safety in Offshore Energy Operations. *Journal of Energy*.
2. Argyle, E. M., Gourley, J. J., Kang, Z., Shebah, R. L. (Under revision: Aug, 2018). Investigating the relationship between eye movements and situation awareness in weather forecasting. *Journal of Applied Ergonomics*.

Peer-reviewed Journal Papers Under Writing

1. Fraga, R. P., Kang, Z, and Mandal, S. Mixed methods approach of visual scanning and verbal protocol analysis. Target journal: *Journal of Human Factors and Ergonomics Society.*
2. Khamaj, A. and Kang, Z. User-centered interface design and evaluation of time-critical services: Application in a mobile weather alert system. Target journal: *Journal of Human Factors and Ergonomics Society.*
3. Jeon, J. and Kang, Z. Towards automatic classification and prediction of complex visual search patterns. Target journal: *IEEE Transactions in Human-Machine Systems.*
4. Kang, Z. and Kim, K. Multimodal analysis of eye movements and keyboard usage of online healthcare systems.
5. Kang, Z. and Kim, K. Analysis user comprehension when interacting with online healthcare systems.

Peer-reviewed Conference Papers (Total: 15)

1. Yoon, D., **Kang, Z.**, Sun, F., and Kim, S. C. (2019). Audiences' attention to Youtube pre-roll advertising: Eye-tracking approach. In *Proceedings of the 2019 Annual Conference of American Academy of Advertising*, Mar. 28-31, Dallas, TX.
2. Kiran, R., Salehi, S., Jeon, J., and **Kang Z.** (2018). Enhancing situation awareness and process safety in offshore drilling operations: Applications of eye-tracking system. In *Proceedings of the 50th Annual Offshore Technology Conference*, Apr. 30-May 3, Houston, TX.
3. Kiran, R., Salehi, S., Jeon, J., & **Kang, Z.** (2018). Real-time eye-tracking system to evaluate and enhance situation awareness and process safety in drilling operations. In *Proceeding of the IADC/SPE Drilling Conference and Exhibition, Society of Petroleum Engineers*, Mar. 6-8. Fort Worth, TX.
4. **Kang, Z.*** (2017). Real time eye movement analysis framework. In *Proceedings of the 2nd International Conference on Bio-engineering for Smart Technologies* (2017 BioSmart), Aug. 30-Sep. 1, Paris, France.
   ***Received Best Presentation Award.**
5. Naeeri, S., and **Kang, Z.** (2017). Analysis of Pilot's Visual Scanning Characteristics under Normal and Extreme Flight Conditions. In *Proceedings of the 6th Annual World Conference of the Society for Industrial and Systems Engineering*, Oct. 12-13, Herndon, VA. 227-233.
6. Khamaj, A., Shehab, R., and **Kang, Z.** (2017). Efficiency and distraction evaluations of motorcycles mounted infotainment systems and controls. In *Proceedings of the 6th Annual World Conference of the Society for Industrial and Systems Engineering*, Oct. 12-13, Herndon, VA. 169-175.
7. Ybarra, V.T., Dadmohammadi, Y., Salehi, S., **Kang, Z.**, Allan, J. N., Ramasubramanian, M., & Cokely, E. T. (2017). Drilling down on cognitive processes: Using process tracing with oil well operators. In *Proceedings of the 2017 European Group of Process Tracing Studies in Judgment and Decision Making*, Jun. 22-24, Galway, Ireland.
8. Yuan, H., Rippetoe, J., Ding, L., **Kang, Z.**, Shehab, R. L., and West. S. G. (2017). Universal Design for Learning in the framework of neuroscience-based education and neuroimaging-based assessment. In *Proceedings of the 2nd International Conference on Bio-engineering for Smart Technologies* (2017 BioSmart), Aug. 30-Sep. 1, Paris, France.
9. Bowden, K. A., Heinselman, P. L., and **Kang, Z.** (2016). Eye-tracking applications to assess impacts of phased array radar data on forecasters' cognitive processes. In *Proceedings of the 32nd Conference*

on Environmental Information Processing Technologies, American meteorological society, New Orleans, LA. 1-6.

10. Mandal, S.*, **Kang, Z.**, and Millan, A. (2016). Data visualization of complex eye movements using directed weighted networks: a case study on a multi-element target tracking task. In *Proceedings of the Human Factors and Ergonomics Society 60th Annual Meeting*, Washington D.C., 106-110.
    ***advisee received Best Student Paper Award from the Aerospace Technical Group**.
11. Mandal, S. and **Kang, Z.** (2015). Eye tracking analysis using different types of Areas of Interest for multi-element moving objects: Results and implications of a pilot study in air traffic control. In *Proceedings of the Human Factors and Ergonomics Society 59th Annual Meeting*, Los Angeles, CA. 1515-1519.
12. **Kang, Z.** and Bass, E.J. (2014). Supporting the eye tracking analysis of multiple moving targets: Design concept and algorithm. *IEEE International Conference on Systems, Man, and Cybernetics*, San Diego, CA, 3191-3196.
13. **Kang, Z.**, Bass, E.J., and Lee, D.W. (2014). Air traffic controllers' visual scanning, aircraft selection, and comparison strategies in support of conflict detection. In *Proceedings of the 58th Annual Meeting of the Human Factors and Ergonomics Society*, Chicago, IL, 77-81.
14. **Kang, Z.** and Bass, E. J. (2013). Computationally tractable approaches to modeling the coverage of weather sensor networks, *IEEE International Conference on Systems, Man, and Cybernetics, Manchester*, England, 4905-4912.
15. **Kang, Z.** and Landry, S. J. (2010). Capturing and analyzing visual groupings of multiple moving targets in an aircraft conflict detection task using eye movements, In *Proceedings of the Human Factors and Ergonomics Society 54th Annual Meeting*, San Francisco, CA, 1906-1910.

Conference Proceedings, Magazines, Posters, and Other Published Manuscripts Not Critically Reviewed (Total: 18)

1. **Kang, Z.**, Shehab, R. L., Ding, L., Yaun, H., West, S. G., Dragoo, M. R., Yeagle, L. N., Fraga, R. P., and Rippetoe, J. (2018). Characterization of air traffic controllers' visual search patterns and control strategies (poster exhibition). In *Proceedings of the 2018 Interservice/ Industry Training, Simulation and Education Conference (I/ITSEC)*, Nov. 26-30, Orlando, FL. **(Poster)**
2. **Kang, Z.**, Shehab, R. L., Ding, L., Yaun, H., West, S. G., Dragoo, M. R., Yeagle, L. N., Fraga, R. P., and Rippetoe, J. (2018). Adaptive learning pedagogy of Universal Design for Learning (UDL) for multimodal training (poster exhibition). In *Proceedings of the 2018 Interservice/ Industry Training, Simulation and Education Conference (I/ITSEC)*, Nov. 26-30, Orlando, FL. **(Poster)**
3. Kang, Z., Jeon, J., Salehi., S., Cokely, E. T., and Ybarra, V. T. (2018). Situation awareness assessment through protocol and eye tracking analyses. In *Proceedings of the 2018 Institute of Industrial and Systems Engineers (IISE) Annual Conference*, May 19-22, Orlando, FL. **(Conference paper)**
4. Mandal, S., and Kang, Z. (2018). Analyzing eye movement data for dynamic scenarios: Information theory and network science based integrated framework. *2nd Annual Gallogly College of Engineering Poster Fair*, University of Oklahoma, Nov. 10, Norman, OK. **(Poster)**
5. Naeeri, S. M., Kang, Z. (2018). Exploring the relationship between pilot's performance and fatigue when interacting with cockpit interfaces. In *Proceedings of the 2018 Institute of Industrial and Systems Engineers (IISE) Annual Conference*, May 19-22, Orlando, FL. **(Conference paper)**
6. Fraga, R. P., Kang, Z. and Mandal, S. (2018). Characterization of air traffic controllers' visual search patterns and control strategies. In *Proceedings of the 2018 ICSTEM, The International Society for Engineers and Researchers*, Jun. 18-19, Seoul, S. Korea. **(Conference paper)**
7. Yeagle, N. L. and Kang, Z. (2018). Universal Design for Learning (UDL) for STEM and higher education: Characterizations and applicability. In *Proceedings of the 2018 ICSTEM, The International Society for Engineers and Researchers*, Jun. 18-19, Seoul, S. Korea. **(Conference paper)**
8. Kang, Z., Dyer, J., West, S. G., Mandal, S., Fraga, R. P., McClung, S., and Egwu, U. K. (2017). Characterization of visual scanning patterns and aircraft control strategies for efficient and effective

training. Poster session in *Solutions for Operational Aviation Research (SOAR) Q2 meeting*, Federal Aviation Administration Center of Excellence, Apr. 3-5, Philadelphia, PA. **(Poster)**
***advisee, Ricardo P. Fraga, received Best Poster Award along with scholarship**

9. Kang, Z., Shehab, R. L., Ding, L., Yaun, H., West, S. G., Dragoo, M. R., Yeagle, L. N., Fraga, R. P., and Rippetoe, J. (2017). Universal Design for Learning and Multimodal Training. Poster session in *Solutions for Operational Aviation Research (SOAR) Q3 meeting*, Federal Aviation Administration Center of Excellence, Apr. 3-5. Philadelphia, PA. **(Poster)**
10. Kang, Z., Dragoo, M.R., Yeagle, L.N., Shehab, R. L., Yuan, H., Ding, L., and West, S. G. (2017). Adaptive learning pedagogy in Universal Design for Learning and multimodal training. In *Proceedings of the 2017 National Training Aircraft Symposium (NTAS)*, Aug. 14-16. Daytona Beach, FL. **(Conference paper)**
11. Kang, Z., Mandal, S., and Dyer, J. (2017). Data visualization approaches in eye tracking to support the learning of air traffic control operations. In *Proceedings of the 2017 National Training Aircraft Symposium (NTAS)*, Aug. 14-16. Daytona Beach, FL. **(Conference paper)**
12. Kang, Z.* and Landry S. J.* (2017). An eye movement analysis algorithm for a multi-element target tracking task: Maximum transition-based agglomerative hierarchical clustering. *IEEE Systems, Man, and Cybernetics Magazine*, *2*, 52-53. **(Magazine)**
    ***Extended abstract published in magazine as a recipient for the Andrew P. Sage Best Transactions Paper Award**.
13. Kang, Z., Shehab, R. L., Ding, L., Yaun, H., West, S. G., Dragoo, M. R., Yeagle, L. N., Fraga, R. P., and Rippetoe, J. (2017). Universal Design for Learning and Multimodal Training. *Solutions for Operational Aviation Research (SOAR) Q4 meeting*, Federal Aviation Administration Center of Excellence, FAA headquarters, Jun. 11-15, Washington D.C. **(Poster)**
14. Kang, Z., Dyer, J., West, S. G., Mandal, S., Fraga, R. P., McClung, S., and Egwu, U. K. (2017) Characterization of visual scanning patterns and aircraft control strategies for efficient and effective training. *Solutions for Operational Aviation Research (SOAR) Q4 meeting*, Federal Aviation Administration Center of Excellence, FAA headquarters, Jun. 11-15, Washington D.C. **(Poster)**

15. Khamaj, A. and Kang, Z. (2017) Usability evaluation of mobile weather alert applications. *1st Annual Gallogly College of Engineering Poster Fair*, University of Oklahoma, Nov. 10, Norman, OK. **(Poster)**
16. Mandal, S. and Kang, Z. (2017, November) Visualizing eye movement data in dynamic object tracking tasks. *1st Annual Gallogly College of Engineering Poster Fair*, University of Oklahoma, Nov. 10, Norman, OK. **(Poster)**
17. Salehi, S., Kiran, R., Jeon, J., Kang, Z., Cokely, E.T., and Ybarra, V.T. (2017) Integrating human factors into petroleum engineering's curriculum: Essential training for students, In *Proceedings of the Society of Petroleum Engineers (SPE) Annual Technical Conference and Exhibition*, Oct. 9-11, San Antonio, TX. **(Conference paper)**
18. Ybarra, V.T., Dadmohammadi, Y., Salehi, S., Kang, Z., Allan, J. N., Ramasubramanian, M., & Cokely, E. T. (2017). Integrating human factors into engineering curriculum: Essential training for students (Poster session), In *Proceedings of the 2017 Offshore Technology Conference*, May 1-4, Houston, TX. **(Conference paper)**

Technical Reports (Total: 7)

1. Kang, Z., Shehab, R. L., Ding, L, Yuan, H., Fraga, R. P., Yeagle, L. N., Alhashim, A., Plata, M. R., Dragoo, M. R. (2019). Universal design for learning and multimodal training. pp. 1-177. Federal Aviation Administration.
2. Salehi, S., Kang, Z., Cokely, E., Muhammed, R. A., Kiran, R., Jiwon, J., Ybarra, V. (2018). Virtual reality offshore operations training infrastructure: Enhancing expert containment, decision making, and risk communications. National Academy of Sciences.

3. Kang, Z., Dyer, J. W., West, S. G., Fraga, R. P., Mandal, S., Egwu, K, and McClung. S. N. (2018). Characterization and application of air traffic controllers' visual search patterns and control strategies for efficient and effective training. pp. 1- 282. Federal Aviation Administration.
4. Kang, Z. and Mandal, S. (2017). Development of design concepts and algorithms in eye tracking research for a multi-element objects tracking task to support human performance analysis. Federal Aviation Administration.
5. Kang, Z. (2017). Development of a mixed methods approach to inform the design of new learning methods for a complex task: An application in multiple objects tracking task. Center for Research Program Development and Enrichment at the University of Oklahoma.
6. Bass, E. J., Kang, Z., Philips, B., Lyons, E., Bajaj, A., Junyent, F., and Beauchamp, R. (2014). Healthcare Systems Integrated Warning Assessment (HIWA) final report. North Central Texas Trauma Regional Advisory Council.
7. Richard, J. P., Kang, Z., and Narisetty, A. K. (2006). SOM v2.0: A simplified optimization model for empty car assignments. Union Pacific Company.

OTHER PRESENTATIONS, PANEL DISCUSSIONS, AND SEMINARS NOT PUBLISHED (Total: 16)

1. Presentation: "Situation awareness assessment through eye movement analysis in Deepwater Horizon operations," *Research meeting with SeaDrill Limited*, Norman, OK. Oct. 31, 2018.
2. Webinar: "Future research topics on visual search and control strategies," *Solutions for Operational Aviation Research (SOAR) Meeting*, Federal Aviation Administration. Jun. 7, 2018.
3. Webinar: "Characterization and application of air traffic controllers' visual search patterns and control strategies for efficient and effective training," *Solutions for Operational Aviation Research (SOAR) Project Closeout Briefing*, Federal Aviation Administration. Apr. 24, 2018.
4. Presentation: "Eye tracking data analytics," *Human Factors and Ergonomics OU Chapter Big Meeting*, Norman, OK, Nov. 1, 2017.
5. Seminar: "Data visualization methods in eye movement research," *OU ISE graduate seminar*, Norman, OK, Oct. 6, 2017.
6. Presentation: "Q4 Characterization of visual scanning patterns and aircraft control strategies for training," *Solutions for Operational Aviation Research (SOAR) Q4 meeting*, Federal Aviation Administration Center of Excellence, FAA headquarters, Washington D.C., Jun. 15, 2017.
7. Presentation: "Q3 Universal Design for Learning (UDL) and multimodal training," *Solutions for Operational Aviation Research (SOAR) Q3 meeting*, Federal Aviation Administration Center of Excellence, Embry-Riddle Aeronautical University, Daytona, FL., 1 Mar. 2017.
8. Presentation: "Q3 Characterization of visual scanning patterns and aircraft control strategies for training," *Solutions for Operational Aviation Research (SOAR) Q3 meeting*, Federal Aviation Administration Center of Excellence, Embry-Riddle Aeronautical University, Daytona, FL., Mar. 1, 2017.
9. Presentation: "Q2 Universal Design for Learning (UDL) and multimodal training," *Solutions for Operational Aviation Research (SOAR) Q2 meeting*, Federal Aviation Administration Center of Excellence, University of Oklahoma, Norman, OK, Dec. 1, 2017.
10. Presentation: "Q2 Characterization of visual scanning patterns and aircraft control strategies for training," *Solutions for Operational Aviation Research (SOAR) Q4 meeting*, Federal Aviation Administration Center of Excellence, University of Oklahoma, Norman, OK, Dec. 1, 2017.
11. Presentation: "Human factors professions and introduction to eye tracking," *Human Factors and Ergonomics Society OU Chapter meeting*, Norman, OK. Sep 15, 2016.

12. Presentation: "Visual scanning analysis" *Research meeting with Laureate Institute for Brain Research*, Norman, OK. Jan 6, 2016.
13. Seminar: "Eye tracking analysis: A demonstration," *Human Factors and Ergonomics Society OU Chapter meeting*, Norman, OK. Mar. 12, 2015.
14. Seminar: "Towards big data analytics: Understanding database management systems," *Special seminar for IIE members and ISE students*, Norman, OK, Mar 9, 2015.
15. Seminar: "Big data analytics: Visual scan path analysis," *Big data analytics meeting*, Center for Research Program Development and Enrichment, Norman, OK, Feb 20, 2015.
16. Seminar: "Eye tracking research: Visual groupings and hierarchical clustering," *OU ISE graduate seminar*, Norman, OK, Oct. 22, 2014.

TEACHING AND ADVISING

Instructor

Advanced Human Factors and Ergonomics / Cognitive Engineering and Decision Making (ISE 5813)*: Methodologies to evaluate human performance and model human decision making processes including human error, signal detection theory, information theory, information processing model, memory, perception, attention, workload, situation awareness, and utility theory. (2015-current)\
**Course name was modified from Cognitive Engineering and Decision Making to Advanced Human Factors and Ergonomics in 2019.*

Class evaluations (out of 5.0):

Year (Sp.)	2015	2016	2017	2018	2019	Mean
Enrollments	12	24	27	27	22	-
Individual	3.4	4.6	4.3	4.8	TBD	4.3
Department	4.3	4.1	4.3	4.2	TBD	4.2
College	4.3	4.2	4.2	4.1	TBD	4.2

**Enrollments have doubled.*

Data Driven Decision Making II (ISE 4853 and ISE 5853)*: Applied statistical methods including mathematical modeling of experiment designs, parametric/non-parametric statistical methods, logistic regression, ANCOVA, database management system, scientific writing, and the use of SAS/R/Microsoft Access. (2015-current)

Class evaluations (out of 5.0)*:

Year (Fa.)	2015	2016	2017	2018	Mean
Enrollments	45	53	77	93	-
Individual	3.2	4.0	4.4	3.9	3.9
Department	4.1	4.3	4.0	4.1	4.1
College	4.1	4.2	4.2	4.1	4.2

**Enrollments have doubled.*

Systems Analysis using Simulation (ISE 4663) and Simulation I (ISE 5663)*: Stochastic simulation theories and applications including manufacturing systems, service systems, human-integrated systems, terminal vs. steady state systems, queuing theory, distributions, non-stationary Poisson process, scheduling, statistical analysis of performance measures, and use of Arena. (2017-current)

Class evaluations (out of 5.0)*:

Year (Fa.)	2017	2018	Mean
Enrollments	74	94	-
Individual	4.0	3.9	4.0
Department	4.0	4.1	4.1
College	4.2	4.1	4.2

Lab evaluations (out of 5.0)*:

Year (Fa.)	2017	2018	Mean
Enrollment	74	94	-
Individual	4.2	4.0	4.1

**Enrollments have doubled.*

Ph.D. Advising (Total: 5)

1. Abdulrahman Khamaj*, Dec 2019.
   Thesis title: Towards an enhanced user experience with smart phone weather alert applications: usability and user-centered design approaches.
2. Elizabeth M. Argyle*, Industrial and Systems Engineering, May 2016.
   Thesis title: Supporting situation awareness and decision making in weather forecasting.
   Co-advised with Dr. Randa L. Shehab.
3. Salem Naeeri, expected Dec 2019.
4. Saptarshi Mandal, expected Dec. 2019.
5. Hunter Kelvgard, expected Dec. 2021.

Abdulrahman is currently a **Tenure-track Assistant Professor at Jazan University in Saudi Arabia.*
Elizabeth is currently a **Research Fellow at the Institute for Aerospace Tech. at the U. of Nottingham.*

M.S. Advising (Total: 7)

1. Elizabeth N. Blanche, Industrial and Systems Eng., Dec 2018.
   Thesis title: Human-computer interface design in human factors: Weapons drop.
2. Jiwon Jeon*, Data Science and Analytics Program, May 2018.
   Thesis title: Analysis of eye tracking data to measure situational awareness in offshore drilling operations.
3. Lucas Cezard, Industrial and Systems Eng., Dec 2017.
   Thesis title: Prediction of user intentions when operating with the cursor.
4. Lucas Mozoni*, Industrial and Systems Eng., Dec 2017.
   Thesis title: Analysis of the relationships between eye tracking information and text comprehension levels in healthcare.
5. Sarah N. McClung*, Electrical and Computer Eng., Aug. 2017.
   Thesis title: Mode reconfigurable bandpass filters with liquid metal enablement.
   Co-advised with Dr. Hjalti H. Sigmarsson..
6. Saptarshi Mandal*. Industrial and Systems Eng., Dec. 2016.
   Thesis title: Analyzing eye tracking data in multielement moving target tracking: Directed weighted network analysis. Dec. 2016.
7. Ricardo P. Fraga*, Industrial and Systems Eng. Dec. 2019 (expected).

**Jiwon is currently a data analyst at a consulting company in Dallas, Texas.*
**Lucas is working at a consulting company in New York, NY.*
Lucas won the 2017 **Best M.S. Thesis award.*
**Sarah is currently a consultant of her family business located in OKC, OK.*
**Saptarshi is currently studying with me as a Ph.D. student.*
Saptarshi won the **Best Student Paper Award from the Aerospace TG in the Proceedings of the 2015*

Human Factors and Ergonomics Society 60th Annual Meeting, Washington D.C.
Saptarshi won **1st place in the Graduate Poster Competition at OU.*
Saptarshi won the 2016 **Best M.S. Thesis award.*
Ricardo won the **Best Poster award during the FAA SOAR meeting, Philadelphia, PA in 2018.*

Ph.D. committee – Thesis option (Total: 5)

1. Katie A. Wilson*, Department of Meteorology, University of Oklahoma, Dec. 2016.
   Thesis title: Forecaster warning decision making with rapidly-updating radar data.
2. Chinedum Peter Ezeakacha, School of Petroleum and Geological Engineering, University of Oklahoma, Dec. 2018.
   Thesis title: Dynamic drilling fluids loss and filtration: impact of dynamic wellbore conditions and wellbore strengthening implications.
3. Austin Graham, Department of Computer Science, University of Oklahoma, (expected) Dec. 2021.
4. Vera Luiza Wendler Bosco, Industrial and Systems Engineering, University of Oklahoma, (expected) Dec. 2021.
5. Aksay Gaur, Department of Computer Science, University of Oklahoma, (expected) Dec. 2021.

**Katie is currently a research associate at the National Weather Service, Norman, OK.*

M.S. committee – Thesis option (Total: 15)

1. Lauren N. Yeagle, Industrial and Systems Eng., OU, Dec. 2018.
2. Bharadwaj Pallerlamudi, Industrial and Systems Eng., OU, May 2017.
3. Karnakar Surabhi, Industrial and Systems Eng., OU, May 2017.
4. Matthew R. Schlegel, Industrial and Systems Eng., OU, May 2017.
5. Naresh Chivukula, Industrial and Systems Eng., OU, May 2017.
6. Nayan Raguluri, Industrial and Systems Eng., OU, May 2017.
7. Enora Maze, Industrial and Systems Eng., OU, Aug. 2016.
8. Sai Krishna Bhavaraju, Industrial and Systems Eng., OU, May 2016.
9. Sidhaarth Krishnamoorthy, Industrial and Systems Eng., OU, Dec. 2016.
10. Somia Tripathy, Industrial and Systems Eng., OU, Dec. 2016.
11. Emily K. Grimes, Industrial and Systems Eng., OU, Dec. 2016.
12. Megan E. Snelling, Industrial and Systems Eng., OU, Dec. 2016.
13. Srujanveer Reddy Kallem, Industrial and Systems Eng., OU, May 2016.
14. Vykram Vijayasekaran, Industrial and Systems Eng., OU, May 2015.
15. Abdulrahman Khmaj, Industrial and Systems Eng., OU, Dec. 2014.

Undergraduate Research Assistants (Total: 6)

1. Jamie Jezier, Industrial and Systems Eng., OU, Fall 2018.
2. Ricardo P. Fraga*, Industrial and Systems Eng., OU, Spring 2018 and Summer 2018.
3. Lauren N. Yeagle*, Industrial and Systems Eng., OU, Spring 2017 and Summer 2017.
4. Mattlyn R. Dragoo*, Industrial and Systems Eng., OU, Spring 2017 and Summer 2017.
5. Brooke Brubaker, Industrial and Systems Eng., OU, Summer 2016.
6. Catherine Ha*, Computer Science, Fall 2015 and Spring 2016.

Lauren and Mattlyn won **1st place at the IISE South Central Regional Conference (held at Texas Tech University in 2017) based on their student project report for my course DDDM II. Title of their presentation at the conference was "Usability analysis of online weather interfaces."*
**Catherine worked as a RA based on my Junior Faculty Fellowship Award.*

PROFESSIONAL SERVICE

Membership

2017-current	Member of Institute of Industrial and Systems Engineers (IISE).
2012-current	Member of IEEE System, Man, and Cybernetics Society (IEEE SMC).
2009-current	Member of Human Factors and Ergonomics Society (HFES).

Offices Held

2018-current	Program Chair of the Education Technical Group in the Human Factors and Ergonomics Society.
2018-current	Editorial Board member of the journal Human Factors and Ergonomics in Manufacturing and Service Industries (Nov. 3 - current)
2018-current	Editorial Board member of the International Journal of Industrial and Operations Research. (Feb. 8 – current)
2018	Guest Academic Editor of the journal PLOS ONE (International Journal of Information Technology and Decision Making).
2016-2018	Program Chair Elect of the Education Technical Group in the Human Factors and Ergonomics Society.
2014-2016	Student Awards and Affairs Officer of the Cognitive Engineering and Decision Making Technical Group in the Human Factors and Ergonomics Society.

Advisor

2014-current	Faculty advisor*, Human Factors and Ergonomics Society OU Chapter. *Dr. Edward Cokely in the Department of Psychology has joined as co-advisor since 2017.

Journal Reviewer

2018-current	Reviewer for the International Journal of Aerospace Psychology
2018-current	Reviewer for the Public Library of Science (PLOS ONE)
2018-current	Reviewer for the International Journal of Environmental Research and Public Health
2016-current	Reviewer for IEEE Transactions on Affective Computing.
2016-current	Reviewer for the journal Applied Ergonomics.
2016-current	Reviewer for the journal Ergonomics.
2016-current	Reviewer for Human Factors & Ergonomics in Mfg. & Service Industries.
2014-current	Reviewer for Human Factors: The Journal of Human Factors and Ergonomics Society.
2013-current	Reviewer for IEEE Transactions on Systems, Man, & Cybernetics.
2013-current	Reviewer for IEEE Transactions on Human-Machine Systems.
2013-current	Reviewer for IEEE International Conference on Systems, Man, & Cybernetics.
2011-current	Reviewer for Human Factors and Ergonomic Society (HFES) Annual Meeting.
2013	Reviewer for International Journal of Human Computer Interaction (HCI).
2011-2012	Reviewer for Journal of Operational Research Society.

Services at the University of Oklahoma

2018-2019	Faculty search committee in ISE Health and Medical Systems.
2017-current	Member of Research Initiatives in Psychology, Human Factors, and Forensics Team to develop research topics and functional requirements for National Environment Simulation and Testing (NEST) facility. Supported by the Office of the Vice President for Research.
2017-2018	ISE director (i.e. chair) search committee member.
2016-current	Advisor of ISE homepage interface design and content management.

2016-2017	Task force team member for SWOT analysis and the development of strategic research focus areas for the "State and Future of the School of the ISE."
2014-current	ISE Graduate Committee.
2014-current	Faculty member to develop courses and reports based on ABET criteria.
2014-current	Advisor of senior undergraduate Capstone projects (Total of 12 projects).

Faculty Development Activities

2018	National Science Foundation (NSF) CAREER Proposal Writing Workshop, NSF Division of Computer, Information Science, and Engineering (CISE) (Apr. 9), NSF Headquarters, Alexandria, VA.
2017	NSF CAREER Proposal Writing Workshop, NSF Division of Civil, Mechanical, Manufacturing and Innovation (CMMI) (Apr. 2-4), Oregon State University, Portland, OR.
2017	Canvas training to enhance student engagement (Jan. 3-6), Norman, OK.
2016	Center of teaching excellence: Faculty learning community - Creating and implementing course videos (Mar. 23 - Apr. 19), Norman. OK.
2015	Faculty Colloquium, Institute of Industrial Engineers (IIE) conference (May 23-24), Nashville, TN.
2014	New faculty seminar series (Sep. 2 - Nov. 18), Norman, OK.

Media Contributions, Appearances, and Interviews

2018 Appeared in headlines of the *Society of Petroleum Engineers* website (www.spe.org) showing collaborative effort among different disciplines.
Author: Saeed Salehi
PIs: Saeed Salehi (Petroleum and Geological Eng.), Ziho Kang (Industrial and Systems Eng.), and Edward Cokely (Psychology).
Occasion: Introducing eye tracking technology for safe oil and gas operations.
Title: Cognitive study of human factors for safe drilling operations: Eye-tracking technology.

2017 Appeared in *IEEE Systems, Man, and Cybernetics Magazine*, 2, 52-53.
Authors: Ziho Kang and Steven Landry
Occasion: Announcement of Andrew P. Sage Best Transactions Paper Award.
Extended Abstract Title: An eye movement analysis algorithm for a multi-element target tracking task: Maximum transition-based agglomerative hierarchical clustering.

GRANTS, COOOPERATIVE AGREEMENTS, CONTRACTS, AND OTHER AWARDS.

Center Awards: Total of $5M

Role	Start	End	Minimum Budget	Sponsor	Title
Institutional Co-PI	8/12/16	8/11/21	$5,000,000	Federal Aviation Administration (FAA)	Center of Excellence for Technical Training and Human Performance (COE TTHP): Solutions for Operational Aviation Research (SOAR)

Other External Awards: $603K of $955K in total

Role	Start	End	Budget	Sponsor	Title
PI	4/1/18	12/31/19	$261,650* (share 100%)	FAA Aviation Research Grants Program (AAQ-610) FAA Cooperative Agreement 692M15184001, Project No. A18-0232.	Analysis of human behaviors and strategies for risk reduction: Application in high fidelity virtual reality simulation of real airports
Co-PI (S. Salehi, PI)	10/1/16	9/30/18	$383,271 (share 30%)	National Academies of Science (NAS) 200007356, Project No. A18-0001.	Virtual reality offshore operations training infrastructure: Enhancing expert containment, decision making, and risk communications
PI	9/9/16	3/8/18	$186,000 (share 57%)	FAA Center of Excellence for TTHP FAA Cooperative Agreement 16-C-TTHP-OK-008 Amend #4, Project No. A17-0161.	Universal Design for Learning (UDL) and multi-modal training
PI	9/13/16	2/12/18	$75,000 (share 95%)	FAA Center of Excellence for TTHP FAA Cooperative Agreement 16-C-TTHP-OK-008 Amend #8, Project No. A17-0162.	Characterization and application of air traffic controllers' visual search patterns and control strategies for efficient and effective training
PI	3/11 /15	12/31/16	$49,545 (share 100%)	FAA Aviation Research Grants Program (AAQ-610) FAA Cooperative Agreement 15-G-006, Project No. 15-P-0009.	Development of design concepts and algorithms in eye tracking research for a multi-element objects tracking and controlling task to support human performance analysis

***** Project was extended with additional funding of $116,010 added to $145,595.

Internal Awards: Total of $10K

Role	Start	End	Budget	Sponsor	Title
PI	8/15/15	5/14/16	$7,000	*Office of the Vice President for Research	Development of a mixed methods approach to inform the design of new learning methods for a complex task
PI	8/15/15	5/14/16	$3,000	Honors Research Assistant Program	Visual scanpath analysis using eye tracking data

**Received as Junior Faculty Fellowship Award*